

Aden Briano

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Education

Rice University – Houston, TX

Master of Science in Electrical & Computer Engineering

Expected Graduation May 2027

Bachelor of Science in Electrical & Computer Engineering

GPA: 3.51 / 4.0, Expected Graduation May 2026

Relevant Coursework: Advanced VLSI Design, High-Performance Computer Architecture, High-Speed System Design, Digital IC Design

Technical Skills

Programming Languages: Python, C, C++, Java, JavaScript, MATLAB, Assembly

Electrical & Computer Engineering Skills: Verilog, SystemVerilog, KiCad, EAGLE, Arduino, ESP32, STM32, MSPM0+, QuestaSim, Magic VLSI, Irsim, Synopsys Design Compiler, Cadence Innovus, LTSpice, HSPICE, Cadence Virtuoso, Vivado

Experience

Systems Engineering Intern | General Dynamics Mission Systems – Scottsdale, AZ May 2025 – Present

- Resolved a CAN communication failure on a **\$100K+** digital control module by tracing it to a misrouted LVDS receiver IC, ensuring correct control voltages for frequency conversion modules, and **preventing a multi-week unit integration delay**
- Authored and redlined test procedures for space-qualified, electrostatic discharge-sensitive digital and RF flight hardware
- Executed functional and performance testing on 14 flight modules**, providing flight-ready hardware that relieved schedule pressure by enabling early unit stackup and feature pass-off on flight hardware

Digital Logic Design TA | Rice University ECE Department – Houston, TX Aug 2025 – Present

- Provide technical guidance to **100+ students** on the design and verification of a single-cycle 16-bit processor

Rover Electrical Team Lead | Rice University Robotics Club – Houston, TX Aug 2024 – Present

- Designed and fabricated a motor controller PCB in KiCad**, integrating DRV8825 stepper drivers and an ESP32 to precisely control five stepper motors on a robotic arm
- Designed a power distribution network and communication systems** to ensure reliable power delivery and telemetry
- Showcased the Rover design to **600+ students** and faculty at the Huff OEDK Engineering Design Showcase

MidWeek Math Trainer | Rice University Math Department – Houston, TX Aug 2024 – May 2025

- Collaborated in a team of 4 to host tailored review sessions twice a week for **200+ students** enrolled in Calculus I/II

Rice Emerging Scholars Program Fellow | Rice University – Houston, TX May 2024 – Aug 2024

- Mentored 50 first-generation STEM freshmen, providing academic coaching and social support during their transition to Rice
- Guided 3 engineering design teams through project scoping, technical planning, and iterative prototyping

Projects

Snake Game ASIC | Verilog, Magic, QuestaSim, Cadence Innovus, Synopsys Design Compiler adenbriano.fyi/snakeasic

- Designed an FSM + Datapath architecture in Verilog** to emulate the classic Snake game in static CMOS Logic
- Verified modules in **QuestaSim**, integrated units, synthesized **50 μ m** logic with **Design Compiler**, and achieved timing closure
- Performed P&R in **Cadence Innovus** and exported for physical layout integration and final simulation in **Magic VLSI**

Adder Design Competition | Cadence Virtuoso adenbriano.fyi/addercomp

- Designed and benchmarked CMOS Full Adders across logic families, achieving a **54% reduction in propagation delay**
- Constructed multiple 32-bit Carry-Square Root Select Adders, reducing delay by **3.3%**, lowering average current draw by **58%**, and optimizing Energy-Delay Product by **61%** through transistor sizing, logical effort analysis, and circuit topology
- Implemented Transmission Gate MUXes and Mirror Full Adders in the fastest CSA design for critical path optimization

Single Cycle 16-Bit Processor | Verilog, FPGA Programming, Vivado github.com/BrianoAden/16-b-Processor

- Developed a single-cycle 16-bit processor in **Verilog** supporting arithmetic and logic instructions
- Simulated and verified functionality using **Vivado waveform viewer**, then programmed onto a **Real Digital Boolean Board**
- Integrated I/O via switches and buttons to execute custom assembly instruction sequences